

Explicit-Boundary BIE/FWI for Scattering: Literature Review, Novelty Assessment, and Research Roadmap

Executive summary

The central conclusion is quite sharp: **generalising an explicitly parameterised single-object Müller/BIE inverse-scattering code to multiple objects, complex topology, or multiple materials is useful and publishable engineering, but is not, by itself, a strong novelty claim.** Multiple dielectric objects have already been treated with spectral boundary-integral solvers and Gauss–Newton inversion; boundary-integral formulations exist for mixtures of penetrable, sound-soft, sound-hard, and absorbing objects; level-set and topological-derivative approaches already address unknown numbers of components and topology change; and, most importantly for the present question, a 2026 preprint explicitly extends the Müller formulation to composite dielectric objects through a global multi-trace construction. ¹

The more promising novelty lies one level deeper: **exploit the fact that the geometry is already represented in Fourier/Laurent coefficients to obtain a genuinely modal, structure-exploiting forward-and-derivative calculus.** The literature contains trigonometric/Fourier Galerkin Müller solvers for arbitrary smooth dielectric resonators, high-order Nyström Müller/transmission schemes, spectral algorithms for dielectric inverse scattering, and shape-calculus theory, but I did **not** find a close precedent combining all of the following in two-dimensional dielectric inverse scattering:

Cartesian Fourier/Laurent geometry + Müller transmission BIE + error-controlled modal operator compression

The absence of a close search hit is not proof of priority, so the novelty confidence here is **medium rather than absolute.** The nearest precedents are trigonometric Galerkin Müller formulations, spectral dielectric inverse-scattering solvers, and recent Fourier-Galerkin operator-compression work in other BIE geometries. ²

My recommended main project is therefore:

Geometry-spectrum-adaptive Fourier-Galerkin Müller operators with analytic or algorithmic shape derivatives for multifrequency inverse scattering.

The key idea is not merely to replace the projected Nyström matrix by another dense $2M \times 2M$ matrix. Instead, test and exploit how the Fourier bandwidth/analyticity of the boundary controls **mode coupling in the Müller operator**. Treat the universal singular part analytically, represent the smooth remainder spectrally, and adaptively truncate/compress mode interactions to a certified tolerance. Construct the corresponding derivative operators $\partial A / \partial c_p$ in the same representation. If successful, this would turn the existing Fourier geometry from a convenient low-dimensional inverse parameterisation into the organising principle of the forward solver itself.

The strongest second direction is:

A differentiable local-scattering-matrix formulation for multiple arbitrary Fourier/Laurent objects, in which each object's Müller solve produces a reusable local response S_j , while interactions are handled by translation/coupling operators.

Multiple-scattering/T-matrix ideas are classical, so **the multiple-object formulation itself is not novel**. The potentially novel contribution would be a boundary-Fourier/Müller local response that supports arbitrary smooth shapes, geometry/material derivatives, selective reuse when only one object changes, and a robust near-touching fallback to a monolithic BIE. Existing multiple-dielectric and inverse-scattering work makes the novelty bar clear. 3

A tempting direction that I would **defer as the main paper** is broad-band reduced-order modelling across frequency and geometry. There is already substantial BEM literature on Krylov-subspace recycling for frequency sweeps and for multiple parameters including **shape and material parameters**, as well as isogeometric BEM work combining frequency decoupling, model-order reduction, and shape optimisation. A new implementation could be valuable, but the novelty margin is thinner unless it is tied to a new Müller/Fourier operator decomposition with rigorous error control. 4

A second important negative conclusion is that **complex topology should not be conflated with an explicit Fourier parameterisation**. A smooth Cartesian Fourier/Laurent curve represents one fixed closed component. Multiple components can be represented by one curve per component, and nested materials by an interface graph, but spontaneous splitting, merging, hole nucleation, or topology change requires an additional mechanism. Level-set and topological-sensitivity methods already cover that regime and have been coupled with BIEs. 5

For a first paper, I would therefore preserve the project's current assumptions—two-dimensional time-harmonic scalar transmission scattering, smooth interfaces, frequency-domain inversion, and explicit Fourier/Laurent boundary variables—and **move deeper into operator structure rather than broader into every possible topology/material configuration**. Multi-object and multi-material capability then becomes an excellent second paper or a major validation/generalisation of the first.

The conclusions below assume the project conventions described in the request: smooth closed boundaries represented by complex Laurent/Fourier series, outgoing two-dimensional Helmholtz Green functions, Müller transmission equations, Kress-type Nyström reference discretisation, Fourier-projected modal systems, and multifrequency receiver-based inversion. Frequency range, exact GPR antenna model, measurement aperture, spatial dimension beyond the current 2D setting, and noise model were not specified; where needed I therefore assume moderate electrical sizes, smooth non-touching interfaces, additive measurement noise, and multiple source/receiver/frequency samples.

Literature landscape and unified vocabulary

The literature divides naturally into four intersecting lines: **boundary-integral forward solvers, shape-differentiable inverse scattering, explicit or low-dimensional geometry representations, and many-query acceleration**.

The forward foundation is mature. Kress and Roach's 1978 Helmholtz transmission formulation is a foundational second-kind transmission BIE; later work has developed single-equation variants, regularised formulations, high-order Nyström schemes, Galerkin methods, and spectral algorithms. Kress's 1991 review is particularly relevant to the project's present nodal benchmark because it directly compares Nyström, collocation, and Galerkin discretisations on smooth two-dimensional boundaries

with special treatment of logarithmic singularities, reporting exponential convergence for analytic data and high efficiency of Nyström discretisation. ⁶

For smooth periodic curves, Kress-type product integration is therefore a very strong reference discretisation rather than a method that needs replacing for accuracy reasons. Modern comparative work on weakly singular Nyström quadratures similarly treats Kress, Alpert, Kapur–Rokhlin and panel-based schemes as high-order alternatives for two-dimensional Helmholtz BIEs. ⁷

There is also direct precedent for the project's proposed **Fourier-Galerkin Müller** direction. Boriskina, Sewell, Benson and Nosich formulated arbitrary-shape two-dimensional dielectric resonator scattering using uniquely solvable second-kind integral equations and a Galerkin discretisation with angular exponentials as global trial/test functions; they analytically extracted the logarithmic singularity and derived closed-form expressions for main parts of the operators. This is probably the single most important prior-art warning against framing “Müller + Fourier Galerkin” by itself as novel. ⁸

For inverse scattering, the intellectual chain runs from the Kirsch–Kress optimisation/integral-equation approach to rigorous domain/shape differentiation and regularised Newton methods. Kirsch and Kress's 1988 work is foundational for integral-equation-based inverse obstacle scattering; Kirsch's 1993 domain-derivative paper, Potthast's 1994 Fréchet differentiability analysis of boundary integral operators, and Hettlich's subsequent work established the analytic machinery underlying Newton/Gauss–Newton shape reconstruction. ⁹

For penetrable media, Hohage's 1998 work applies an iteratively regularised Gauss–Newton method to the Helmholtz transmission inverse problem, while more recent spectral dielectric solvers couple second-kind BIEs, explicit boundary parametrisations and shape derivatives with regularised Gauss–Newton reconstruction. ¹⁰

The important implication is that **shape derivatives themselves are not novel**. The opportunity is to make them *computationally native* to the chosen Fourier/Laurent representation. In other words, instead of deriving a continuous shape derivative and then rediscratising it on nodal geometry, one can ask whether

$$c_p \longmapsto A_M(\mathbf{c}, k)$$

admits an efficiently evaluable coefficient-space derivative

$$\frac{\partial A_M}{\partial c_p},$$

with the same singularity separation and modal compression as A_M itself. The theoretical foundations for differentiating scattering operators are established, including penetrable electromagnetic interfaces and boundary integral operators. ¹¹

Explicit geometry representations are much less uniform across the literature. “Fourier boundary” can mean several quite different things, and this matters for novelty claims.

Literature term	Unified term used here	Important distinction
Trigonometric polynomial, Fourier curve, band-limited shape	Explicit Fourier boundary	Coordinates or normal displacement are finite Fourier series.
Laurent polynomial/series $z(t) = \sum c_m e^{imt}$	Cartesian Fourier/Laurent boundary	Both Cartesian coordinates are encoded simultaneously in complex coefficients; does not require star-shapedness.
Polar Fourier, radial graph $r(\theta)$	Star-shaped Fourier boundary	Requires a centre from which the boundary is single-valued.
B-spline / cubic spline	Spline boundary	Local support; fixed component topology.
NURBS / isogeometric BEM	CAD-explicit boundary	Same functions describe geometry and often solution basis; especially mature in shape optimisation.
Level set	Implicit interface representation	Mathematically not an explicit boundary parameterisation, although the level-set function can itself have finitely many coefficients.
Parametric level set / B-spline level set	Coefficient-parametrised implicit interface	Supports topology changes more naturally than a single explicit closed curve.
RBF level set / neural implicit	Implicit basis boundary	Boundary is a zero set; geometry parameters are RBF/network coefficients rather than boundary coordinates.
Domain derivative, material derivative, shape derivative	Shape derivative	Different technical constructions for differentiating with respect to a moving interface.
Multiple obstacle / multiple scatterer / multi-component	Multi-object geometry	Separate disjoint components; each may have its own material.
Composite, multidomain, multi-trace, multilayer	Multi-material interface network	Regions and interfaces form an adjacency graph; nested or touching materials are possible.
T-matrix / transition matrix / scattering matrix	Local response operator	Maps incident modal coefficients around one object to outgoing modal coefficients.
Model-order reduction / reduced basis / Krylov recycling	Solution-space reduction	Compresses the family of solved states.
Affine decomposition / kernel expansion / frequency separation	Operator-space reduction	Compresses $A(k, \mathbf{c})$ itself.

Literature term	Unified term used here	Important distinction
H-matrix / skeletonisation / recursive skeletonisation / FMM-LU	Data-sparse BIE acceleration	Compresses far interactions or factors dense BIE matrices.

Spline and CAD-explicit parameterisations are already well represented in adjacent inverse and shape-optimisation fields. For example, cubic splines have been used to represent buried dielectric-cylinder contours in time-domain inverse scattering, while NURBS-based isogeometric BEM is well developed for electromagnetic scattering and Helmholtz shape optimisation. ¹²

Level-set+BIE work is also mature enough that “allow topology change” is not a fresh contribution. He, Kindermann and Sini coupled level-set updates with boundary-integral forward/adjoint solves for simultaneous reconstruction of shape and impedance; more recent inverse-scattering work has developed accelerated level-set schemes, and topological derivatives have been demonstrated for detecting multiple electromagnetic scatterers without knowing their number beforehand. ⁵

For GPR/subsurface applications specifically, the screened literature contains boundary-integral or boundary-domain approaches for buried impedance/dielectric configurations, cross-borehole inverse scattering, and buried-profile reconstruction beneath dielectric rough surfaces. These establish application relevance, but none of the sources screened duplicates the exact combination of Cartesian Fourier/Laurent geometry, Müller modal discretisation, and derivative-aware multifrequency inversion proposed here. ¹³

Candidate research directions and Pareto comparison

Scores below are **decision scores**, not bibliometric facts. Novelty reflects how much space appears to remain after the closest precedents found; implementation and validation difficulty use **10 = hardest**. “Theory” reflects the opportunity for a mathematical contribution rather than merely software development.

Candidate direction	Novelty	Impl. difficulty	Validation difficulty	Theory interest	Practical payoff	Evidence confidence	Verdict
Geometry-spectrum-adaptive Fourier-Galerkin Müller operator + coefficient-space shape derivatives	8.5	6	6	9.5	8	Medium	Recommended main direction

Candidate direction	Novelty	Impl. difficulty	Validation difficulty	Theory interest	Practical payoff	Evidence confidence	Verdict
Multi-object local-response Müller calculus with derivative/reuse machinery	7.5	7	7	8	9	Medium-high	Strongest alternative/generalisation
Geometry/frequency-separated Müller assembly with derivative-aware operator compression	5.5	6	6	7.5	8	High	Good component, weak standalone novelty
Selective factorisation/preconditioner updates across FWI geometry steps	6	7	6	6.5	9	Medium-high	Strong computational paper if speedups are large
Broadband reduced-basis/Krylov Müller solver under changing geometry/material	4.5	7	7	6	8	High	Tempting but crowded; defer
Explicit multi-material/multi-trace Müller with interface graph	4.5	8	8	8	8	High	Important, but 2026 prior art sharply lowers novelty
Topology-changing level-set/RBF interface inversion with BIE	3.5	9	9	7	7	High	Established field; large backend change

Candidate direction	Novelty	Impl. difficulty	Validation difficulty	Theory interest	Practical payoff	Evidence confidence	Verdict
Time-domain/ CQ replacement of frequency- domain BIE backend	4	10	9	8	7	Medium-high	High risk relative to present project

The first candidate is attractive because it asks a **specific structural question that the present code can answer almost immediately**:

Does the modal Müller matrix inherit predictable compressibility from the Fourier spectrum and analyticity of the boundary?

Let

$$A_{mn}^{\alpha\beta}(\mathbf{c}, k) = \frac{1}{(2\pi)^2} \int_0^{2\pi} \int_0^{2\pi} e^{-imt} K_{\alpha\beta}(t, s; \mathbf{c}, k) e^{ins} ds dt,$$

where α, β index the two Müller traces. After a standard logarithmic singularity split,

$$K(t, s) = a(t, s) \ell(t - s) + b(t, s), \quad \ell(\theta) = \log\left(4 \sin^2 \frac{\theta}{2}\right),$$

the singular factor ℓ has explicit Fourier coefficients, whereas a and b are smooth periodic functions for a smooth boundary. Existing trigonometric-Galerkin Müller work already exploits this idea to isolate the singular piece. ⁸

The new question is whether one can go beyond “spectral accuracy” to establish and exploit a relation of the form

$$|A_{mn}| \lesssim F(|m - n|; \text{geometry spectrum}, k),$$

or an a posteriori analogue strong enough to drive adaptive mode-coupling truncation. Recent Fourier-Galerkin BIE work on torus-like surfaces explicitly exploits decay patterns in representation matrices to obtain sparse approximations, so there is an adjacent precedent for the *kind* of analysis, but not an identified close version for two-dimensional Müller inverse scattering with evolving Fourier geometry.

¹⁴

The second candidate uses a block response structure. For object j , let a_j and b_j denote local incoming and outgoing modal coefficients and define a local response

$$b_j = S_j(\mathbf{c}_j, k, \epsilon_j) a_j.$$

Interactions satisfy approximately/exactly, depending on basis and configuration,

$$a_j = a_j^{\text{inc}} + \sum_{\ell / = j} U_{j\ell} b_\ell,$$

hence

$$(I - SU)b = S a^{\text{inc}}.$$

This is classical multiple-scattering/T-matrix structure, so the novelty cannot be “we use a scattering matrix for several objects.” Multiple dielectric cylinders and multiple dielectric inverse problems are already represented in the literature. ¹⁵

The plausible contribution is instead:

$$S_j = \text{arbitrary-shape Fourier/Laurent Müller response,}$$

together with

$$\partial_{\mathbf{c}_j} S_j, \quad \partial_{\epsilon_j} S_j,$$

and rules for reusing every S_ℓ whose object did not change during an optimisation step. That would make “multi-object generalisation” a **computational architecture for inverse scattering**, not simply an extra block row in a BIE.

The third candidate decomposes frequency dependence:

$$A(\mathbf{c}, k) \approx \sum_{r=1}^R \theta_r(k) A_r(\mathbf{c}),$$

or more ambitiously

$$A(\mathbf{c}, k) \approx \sum_{r=1}^R \theta_r(k) \phi_r(\mathbf{c}) A_r,$$

with compatible derivative approximations. This is highly relevant but already crowded. Acoustic BEM literature uses Green-function series expansions and Krylov recycling for frequency sweeps; multiparameter MOR explicitly includes source position, material and **shape parameterisations**; and isogeometric electromagnetic BEM has combined frequency-decoupling, reduced-order modelling and CAD-integrated shape optimisation. ⁴

Thus a paper titled simply “reduced-order BEM for multifrequency shape optimisation” would face strong prior art. It becomes more defensible only if the reduction follows from a genuinely new Fourier-Müller operator representation and includes a uniform error story for both forward fields and inverse gradients.

Detailed assessment of the leading directions

Recommended: geometry-spectrum-adaptive Fourier-Galerkin Müller.

The central contribution should not be “direct Laurent coefficient assembly” in the strongest literal sense. A pure symbolic coefficient pipeline is likely to encounter **bandwidth explosion** because the Helmholtz kernel contains

$$H_0^{(1)}(k|\gamma(t) - \gamma(s)|),$$

while normals, Jacobians, and hypersingular/derivative terms involve nonlinear functions of γ' and $|\gamma'|$. Even if γ has only M_g Fourier modes, these nonlinear compositions generally do not preserve that bandwidth. The smooth part of the bivariate kernel can therefore become spectrally dense, especially as electrical size increases.

The safer and, in my view, stronger algorithmic design is **hybrid analytic/spectral assembly**:

$$A = A_{\text{universal sing}} + A_{\text{smooth}},$$

where the universal singular symbols are treated analytically using known Fourier coefficients, while the smooth two-variable factor is sampled on an oversampled tensor grid, transformed with FFTs, and compressed/truncated according to an error criterion. This retains the exact Fourier-Galerkin interpretation while avoiding the fragility of trying to express every Hankel composition as a closed coefficient formula. Trigonometric Müller Galerkin precedent supports the singularity-extraction part; the novelty would lie in adaptive modal structure, geometry-driven compression and derivative reuse.

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For a boundary coefficient c_p , derive or automatically differentiate

$$\frac{\partial A_{mn}}{\partial c_p} = \left\langle e_m, \frac{\partial A}{\partial c_p} e_n \right\rangle.$$

The continuous validity of such differentiation is supported by the established shape-calculus literature; the computational novelty is to show that $\partial A / \partial c_p$ has the **same exploitable modal structure** as A .

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A particularly attractive theorem/result would be one of the following forms:

$$\|A - A_B\| \leq C e^{-\sigma B}$$

for analytic boundaries and an appropriate band/coupling truncation A_B , or a posteriori:

$$\|A - A_{\mathcal{I}}\| \leq C \left(\sum_{(m,n) \notin \mathcal{I}} |A_{mn}|^2 \right)^{1/2},$$

with a practical estimator linking the required interaction set $\mathcal{I}$ to geometry bandwidth, electrical size and desired tolerance.

That would distinguish the work from generic spectral convergence: it would explain **why many mode pairs can be omitted and when they cannot**.

The principal failure mode is straightforward: at high kD , on strongly oscillatory/high-curvature boundaries, or near resonant configurations, the modal operator may be too dense for meaningful truncation. That is why the first experiment should test the hypothesis *before* implementing a new assembler.

Strongest alternative: differentiable multi-object local-response Müller method.

Multiple objects are a natural extension of the existing explicit-boundary code. For J disjoint inclusions, a monolithic Müller system has block form

$$\begin{bmatrix} A_{11} & A_{12} & \cdots \\ A_{21} & A_{22} & \cdots \\ \vdots & & \ddots \end{bmatrix} x = b,$$

where self blocks contain singular quadrature and cross blocks are smooth.

That extension is straightforward but not novel. Multiple dielectric scattering, spectral inverse reconstruction of multiple dielectric obstacles, and heterogeneous collections of penetrable and impenetrable objects already exist. ³

A stronger formulation eliminates repeated local solves into reusable response operators. For each object,

$$S_j = R_j A_{jj}^{-1} B_j,$$

where B_j maps an incident modal field to Müller boundary data and R_j maps solved traces to outgoing modal coefficients. If only object p changes at one Gauss–Newton step, then ideally only S_p must be recomputed; translations among unchanged objects can be retained except where positions move.

Differentiating the global system gives, schematically,

$$(I - SU) \delta b = \delta S (a^{\text{inc}} + Ub) + S \delta U b + S \delta a^{\text{inc}}.$$

This gives a clean way to separate **shape**, **material**, and **position** derivatives.

The most publishable version would cover:

1. arbitrary smooth Cartesian-Fourier objects rather than circles/spheres;
2. transmission rather than only sound-soft obstacles;
3. exact comparison against the monolithic Müller system;
4. analytic/adjoint derivatives through S_j ;
5. selective reuse across optimisation iterations;
6. a near-touching strategy.

The last item matters. Multipole/translation bases lose efficiency as objects approach one another because more local modes are needed to resolve strong near interactions. A robust hybrid could treat close clusters with a monolithic BIE while coupling well-separated clusters through local response matrices. That type of **adaptive algebraic decomposition** is more interesting than merely adding more objects.

For multi-materials, two cases must be kept distinct. Separate homogeneous objects are easy: each S_j just has its own $k_{i,j}$ or material parameters. Nested/coated/composite objects require an **interface adjacency graph**, since one boundary separates two specific regions and the unknown traces are coupled region-by-region. Classical multidomain/multi-trace BIE machinery addresses this, and the 2026 Le–Cool’s preprint now explicitly describes a multi-trace Müller BIE for composite dielectric objects. That makes “Müller for composite materials” a weak novelty claim going forward. ¹⁸

Tempting direction to defer: broad-band MOR over frequency and geometry.

Many-query acceleration is clearly valuable because inverse scattering requires repeated forward/adjoint solves. A reduced model might use

$$x(k, \mathbf{c}) \approx Vz(k, \mathbf{c}), \quad V \in \mathbb{C}^{N \times r}, \quad r \ll N,$$

or an operator approximation

$$A(k, \mathbf{c}) \approx \sum_r \theta_r(k, \mathbf{c}) A_r.$$

However, acoustic BEM Krylov recycling already targets large frequency sweeps, and a 2022 multiparameter extension explicitly demonstrates source, material and shape parameterisations. ¹⁹

Likewise, recent isogeometric BEM work combines NURBS geometry, frequency-decoupled kernels, second-order Arnoldi MOR and electromagnetic shape optimisation. ²⁰

The novelty gap is therefore narrow. A successful project would need to demonstrate something stronger than “ROM makes repeated solves faster,” for example:

$$\sup_{(k, \mathbf{c}) \in \mathcal{P}} \frac{\|F_{\text{ROM}} - F\|}{\|F\|} \leq \varepsilon$$

and

$$\sup_{(k, \mathbf{c}) \in \mathcal{P}} \frac{\|\nabla_{\mathbf{c}} J_{\text{ROM}} - \nabla_{\mathbf{c}} J\|}{\|\nabla_{\mathbf{c}} J\|} \leq \varepsilon_g,$$

with an online cost reduction large enough to outweigh offline training. Gradient fidelity is the important metric for FWI; field error alone is insufficient.

I would therefore use frequency/geometry reduction as a **second-stage accelerator built on the recommended modal operator**, not as the initial novelty claim.

Novelty gaps, decisive experiments, and stop criteria

The central prior-art boundary can be stated compactly.

Area	Already established	Contribution still plausibly novel
Müller + Fourier/trigonometric Galerkin	Arbitrary-shape 2D dielectric scattering with angular-exponential Galerkin and analytic singularity extraction. ⁸	Geometry-spectrum-dependent adaptive coupling/compression , inverse derivatives, and FWI-oriented error control.
Shape derivatives	Rigorous domain/Fréchet derivatives for acoustic and EM scattering and BIE operators. ²¹	Coefficient-space derivative assembly that reuses the same compressed Fourier operator machinery.
Spectral inverse dielectric BIE	Spectral second-kind BIE + IRGNM for dielectric obstacle recovery. ²²	Explicit Cartesian/Laurent geometry without a star-shaped restriction plus structured modal derivative calculus.

Area	Already established	Contribution still plausibly novel
Multiple objects	Direct/inverse multiple dielectric scattering and heterogeneous multi-object BIEs. ²³	Arbitrary-shape local response + derivative/reuse architecture , preferably with cluster treatment for close objects.
Composite/multi-material Müller	Multi-trace composite Müller now exists in 2026 preprint literature. ¹⁸	Inverse/material derivative machinery tied to low-dimensional explicit interface parameters and reusable regional blocks.
Frequency-reduced BEM	Krylov recycling, multiparameter BEM MOR, frequency-decoupled IGABEM. ⁴	Uniformly accurate derivative-aware reduction specifically exploiting Müller/Fourier structure.
Topology change	Level-set+BIE and topological derivatives already detect/reconstruct multiple components. ²⁴	Little novelty unless a substantially new BIE-compatible topology mechanism or complexity result is introduced.

The **first decisive experiment** should require almost no new solver.

Take the existing Kress Nyström matrix A_N and the current DFT projector P , and form the diagnostic modal matrix

$$\tilde{A}_M = P^* A_N P.$$

Do this **only as a diagnostic**, not as the final claimed Galerkin discretisation. For circles, ellipses, kites, peanut-like non-star-shaped curves, and one deliberately high-Fourier-content boundary, measure

$$E_B = \frac{\|\tilde{A}_M - \mathcal{T}_B(\tilde{A}_M)\|_F}{\|\tilde{A}_M\|_F},$$

where $\mathcal{T}_B$ keeps modal interactions within a proposed geometry-dependent interaction set.

Sweep at least three electrical sizes, for example roughly $kD = 2, 10, 30$. Compare:

modal matrix truncation error, boundary-trace error, receiver-field error, shape-gradient error.

A particularly useful plot is $|A_{mn}|$ as an image, overlaid with the spectral bandwidth of $\gamma(t)$.

Success criterion: there is a stable rule tying coupling bandwidth to geometry spectrum and kD , with receiver and gradient accuracy preserved while retaining materially fewer than $O(M^2)$ significant modal interactions.

Kill criterion: for the geometries/frequencies relevant to the intended application, accurate matrices are effectively dense and their coupling structure changes unpredictably under modest shape perturbations. In that case, pure modal compression should be abandoned and the effort redirected to local-response reuse or hierarchical dense linear algebra.

The **second experiment** should test multi-object local responses before a full inverse implementation.

For two circles, construct local S_1, S_2 and compare against an analytic cylindrical-wave/Mie benchmark and against the monolithic Müller Nyström solve. Then replace one circle by an ellipse and one by a Fourier-defined non-star-shaped object. Vary separation from several wavelengths down toward the near-touching regime.

Measure

$$\epsilon_{\text{field}} = \frac{\|u_{\text{local}} - u_{\text{monolithic}}\|}{\|u_{\text{monolithic}}\|},$$

required local modal order, assembly time, factorisation time, and repeated-source cost.

Then perturb only object 1 and time:

rebuild all versus rebuild only S_1 .

Success criterion: local response orders remain much smaller than full global nodal dimensions over the intended separation range, and selective reuse gives a compelling many-source or optimisation-step speedup.

Kill criterion: useful accuracy requires a local multipole order comparable to the full boundary discretisation over ordinary target separations, or near-field corrections become so elaborate that a monolithic BIE is simpler and equally fast.

The **third experiment** should test broad-band low rank without writing a ROM.

Generate projected/modal matrices at 20–40 frequencies,

$$A(k_1), \dots, A(k_Q),$$

vectorise representative operator blocks or evaluate them on a common set of probe vectors, and perform SVD/rational interpolation. Repeat for several nearby geometries.

Test separately:

operator rank in k , solution-manifold rank in k , joint $(k, \mathbf{c})$ rank.

The distinction matters: a solution family can be low dimensional even when the operator family is not.

Success criterion: a rank of perhaps 5–10 reproduces held-out frequencies at forward and derivative tolerances relevant to inversion over a genuinely useful band.

Kill criterion: rank grows nearly linearly with sampled frequency count, or modest geometry changes invalidate the basis, particularly around resonances. In that case, use frequency continuation and factorisation/preconditioner recycling instead of a global ROM.

A fourth, very inexpensive diagnostic would test **preconditioner reuse**. Use the LU/preconditioner at $(k_n, \mathbf{c}_n)$ for solves at $(k_{n+1}, \mathbf{c}_n)$ and $(k_n, \mathbf{c}_{n+1})$, recording GMRES iterations versus Δk and $\|\Delta \mathbf{c}\|$. Factorisation updating for evolving boundary-integral geometries has precedent, including local geometric perturbations in multiply connected domains, so any paper here should emphasise Helmholtz/Müller inverse-scattering consequences rather than factorisation updating itself. 25

Recommended research path and visual map

I recommend a staged programme in which each step produces evidence before the next backend is written.

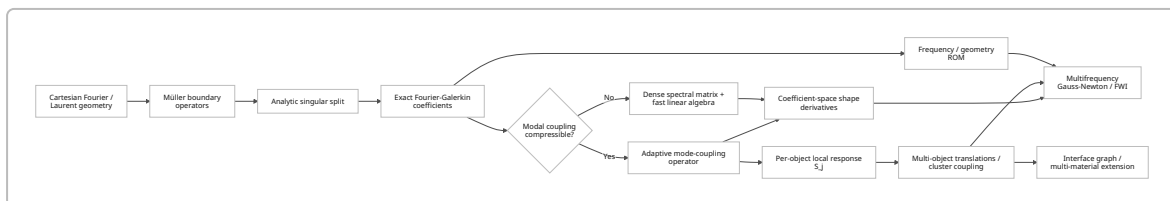
The first stage should **audit structure using the existing code only**. Compute projected modal matrices, their singular/coupling patterns, derivative matrices, and frequency variation. This directly decides whether the strongest proposed novelty—geometry-induced modal compressibility—is real in the application regime.

The second stage should implement an **exact Fourier-Galerkin reference** for the Müller equations, but initially without aggressive direct coefficient algebra. Analytically treat the universal logarithmic singular component and evaluate smooth coefficients by heavily oversampled quadrature/FFT. This cleanly removes projected-Nyström aliasing from the study while keeping implementation risk controlled. The existing Kress Nyström solver remains the reference. The established success of both high-order Nyström and trigonometric Galerkin schemes makes this a well-grounded validation architecture. ²⁶

The third stage should introduce **adaptive modal compression and derivative assembly**. Only after the empirical decay law is known should you decide whether the smooth coefficients warrant direct Laurent convolution algebra, sparse banded representations, low-rank blocks, FFT application, or a mixture of them.

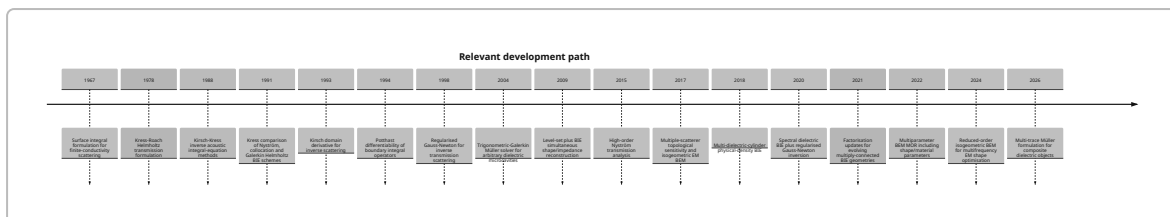
The fourth stage should expose a local object response S_j . That converts the single-object work into the proposed multi-object formulation without changing the local Müller mathematics.

The fifth stage can then choose between multi-material interface graphs and broadband MOR according to application need. The existence of contemporary multi-trace composite Müller work means the multi-material paper must focus on inverse/derivative/reuse structure rather than forward well-posedness alone. ¹⁸



This relationship is consistent with the literature: spectral/second-kind BIEs and Gauss–Newton inversion are established, while multiple-scattering, reduced-order and factorisation-reuse techniques can be attached as outer algebraic layers. ²⁷

A useful development timeline is:



The timeline is supported by the primary and near-primary literature reviewed below. 28

A practical paper sequence would therefore be:

Main paper: Fourier/Laurent geometry → exact Fourier-Galerkin Müller → geometry-spectrum adaptive compression → shape derivatives → multifrequency inversion.

Follow-on paper: arbitrary-shape local response matrices → multiple-object coupling → selective derivative/factorisation reuse → multi-target FWI.

Later extension: nested/composite materials through an interface graph/multi-trace formulation, adding material-variable inversion.

I would **not** initially combine topology change, 3D Maxwell, broadband ROM, multiple objects and multiple materials into one framework. That would increase implementation risk while making it harder to identify what is actually novel.

Ranked literature review

The table below contains the **materially relevant literature retained from the search**, ranked for this specific project rather than by general citation importance. A relevance score of 10 means “directly defines the novelty boundary for the proposed work.”

Rank	Citation / year	Short relevance summary	Relevance	Why important for novelty
1	Boriskina, Sewell, Benson & Nosich, “Accurate simulation of two-dimensional optical microcavities with uniquely solvable boundary integral equations and trigonometric Galerkin discretization,” 2004. 8	Arbitrary-shape 2D dielectric second-kind BIE with global angular-exponential Galerkin basis and analytic singular extraction.	10	Closest direct precedent to Fourier-Galerkin Müller; prevents claiming that basic combination as novel.
2	Le Louër, “A spectrally accurate method for the direct and inverse scattering problems by multiple 3D dielectric obstacles,” 2017/2018. 29	Spectral second-kind BIE and regularised Gauss–Newton reconstruction for multiple piecewise-homogeneous dielectric obstacles.	10	Strongest warning that “multi-object dielectric inverse BIE” alone is established.

Rank	Citation / year	Short relevance summary	Relevance	Why important for novelty
3	Hohage & Le Louër, spectral dielectric obstacle inverse scattering, 2020. ²²	Second-kind BIE, spectral discretisation, shape differentiation, adjoint and iteratively regularised Gauss–Newton for dielectric shape recovery.	10	Very close conceptual pipeline to the proposed inverse solver, especially in 3D.
4	Panagiotopoulos, Desmet & Deckers, multiparameter MOR for acoustic BEM, 2022. ³⁰	Krylov recycling/ reduced bases for BEM systems with frequency plus source, material and shape parameters.	9.8	Strong prior art against generic “frequency + geometry reduced BEM” claims.
5	Chen et al., reduced-order isogeometric BEM for multifrequency electromagnetic shape optimisation, 2024. ²⁰	Frequency decoupling, SOAR MOR, NURBS geometry and EM shape optimisation.	9.8	Extremely close to a geometry/frequency separated-operator research pitch.
6	Le & Cools, “Multi-Trace Müller Boundary Integral Equation for Electromagnetic Scattering by Composite Objects,” 2026 preprint. ¹⁸	Extends Müller to composite dielectric structures through a global multi-trace formulation.	9.8	Current prior art materially lowers novelty of a plain multi-material Müller extension.
7	Kress & Roach, “Transmission problems for the Helmholtz equation,” 1978. ³¹	Foundational Helmholtz transmission formulation underlying many well-conditioned second-kind BIEs.	9.7	Mathematical foundation for Müller/Kress–Roach-type transmission solvers.
8	Kress, “Boundary integral equations in time-harmonic acoustic scattering,” 1991. ³²	Compares Nyström, collocation and Galerkin schemes on smooth 2D boundaries with singular quadrature treatment.	9.7	Core reference for nodal Kress Nyström versus Fourier-Galerkin design choices.

Rank	Citation / year	Short relevance summary	Relevance	Why important for novelty
9	High-order Nyström discretisations for 2D Helmholtz transmission problems, 2015/2016. ³³	Rigorous fully discrete Nyström analysis for several well-conditioned dielectric transmission formulations.	9.6	Strong modern reference for the existing nodal benchmark.
10	Kirsch, "The domain derivative and two applications in inverse scattering theory," 1993. ³⁴	Foundational domain-derivative theory for inverse scattering.	9.5	Establishes that continuous shape differentiation is classical.
11	Potthast, "Fréchet differentiability of boundary integral operators in inverse acoustic scattering," 1994. ³⁵	Directly analyses boundary-integral operators under boundary perturbation.	9.5	Critical theoretical base for derivative-aware operator assembly.
12	Hohage, "A Newton-type method for a transmission problem in inverse scattering," 1998. ³⁶	Iteratively regularised Gauss–Newton applied to nonlinear, severely ill-posed Helmholtz transmission inversion.	9.4	Establishes the basic inverse-transmission optimisation architecture.
13	Kirsch, Kress et al., "Two methods for solving the inverse acoustic scattering problem," 1988. ³⁷	Foundational integral-equation optimisation methodology for inverse obstacle scattering.	9.3	Historical origin of much explicit-boundary BIE inversion.
14	Ryan & Damle, "Parallel Skeletonization for Integral Equations in Evolving Multiply-Connected Domains," 2021. ²⁵	Updates hierarchical BIE factorizations efficiently after local geometry changes; demonstrated in shape optimisation.	9.2	Closest structural prior art for geometry-step factorisation reuse.
15	Helsing & Karlsson, physical-density integral equations for multi-dielectric cylinders, 2018. ³⁸	High-accuracy BIE framework for multiple dielectric interfaces, including difficult near evaluations.	9.1	Strong forward precedent for multiple arbitrary dielectric boundaries.

Rank	Citation / year	Short relevance summary	Relevance	Why important for novelty
16	Le Louër & Rapún, mixed-nature multiple-scattering BIE/topological method, 2022. ³⁹	Boundary-integral system for simultaneous penetrable, soft, hard and absorbing 3D scatterers.	9.0	Shows heterogeneous multi-object forward BIEs are already broad.
17	Costabel & Le Louër, shape derivatives of EM boundary-integral operators, 2010–2011. ⁴⁰	Infinite shape differentiability of standard electromagnetic BIE operators and penetrable scattering solutions.	9.0	Strong theoretical base for material/interface derivative work.
18	He, Kindermann & Sini, “Reconstruction of shapes and impedance functions using few far-field measurements,” 2009. ⁴¹	BIE forward/adjoint solves with level-set shape updates and simultaneous impedance reconstruction.	8.9	Demonstrates topology-flexible BIE inversion and simultaneous geometry/boundary-property recovery.
19	Askham & Borges, reconstructing shape and material parameters of dissipative obstacles, 2024. ⁴²	Multifrequency optimisation recovering both geometry and material-related impedance parameters.	8.9	Recent evidence that simultaneous shape/material inversion is active prior art.
20	Borges & Rachh, multifrequency inverse obstacle scattering with unknown impedance, 2021. ⁴³	Band-limited shape/impedance representation with recursive linearisation across frequency.	8.8	Strong precedent for band-limited geometry plus multifrequency inversion.
21	Le Louër & Rapún, topological sensitivity for multiple electromagnetic scatterers, 2017. ⁴⁴	Detects multiple scatterers without prior knowledge of number, size or location.	8.7	Makes “unknown number of objects/topology” a crowded novelty direction.
22	Accelerated level-set inverse scattering, Audibert, Haddar & Liu, 2022. ⁴⁵	PDE-constrained shape optimisation with level-set geometry and accelerated updates.	8.6	Modern competitor for complex/topology-changing shapes.

Rank	Citation / year	Short relevance summary	Relevance	Why important for novelty
23	Simpson et al., isogeometric BEM for electromagnetic scattering, 2017. ⁴⁶	NURBS geometry, compatible B-spline Galerkin discretisation and hierarchical-matrix acceleration.	8.5	Important CAD-explicit alternative to Fourier/Laurent curves.
24	Takahashi, Sato, Isakari & Matsumoto, NURBS isogeometric BEM + adjoint shape optimisation, 2021. ⁴⁷	Uses NURBS control points directly as 3D acoustic design variables with adjoint BEM.	8.5	Shows explicit CAD geometry + BEM + shape derivative is established.
25	Liu et al., isogeometric BEM for dielectric EM target shape optimisation, 2024. ⁴⁸	NURBS-based frequency-domain BEM for 2D dielectric scattering shape optimisation.	8.5	Particularly relevant competitor to explicit-boundary dielectric design.
26	Panagiotopoulos, Deckers & Desmet, BEM Krylov-recycling MOR, 2020. ⁴⁹	Green-kernel expansion plus reduced basis for frequency-sweep acoustic BEM.	8.4	Key frequency-reuse reference.
27	Martinsson & Rokhlin, fast direct solver for 2D BIEs, 2005. ⁵⁰	Hierarchical compressed direct factorisation particularly efficient for multiple right-hand sides.	8.4	Foundational route for fast repeated BIE solves independent of Fourier structure.
28	Sushnikova, Greengard, O'Neil & Rachh, FMM-LU, 2022/2023. ⁵¹	Hierarchical direct BIE factorisation with fast inverse application for low/moderate-frequency Helmholtz.	8.3	Relevant benchmark against any proposed custom modal acceleration.
29	Pinto, Aylwin & Jerez-Hanckes, spectral Galerkin method for multilayer Helmholtz transmission, 2021. ⁵²	Fourier-parametrised spectral Galerkin BIEs for multilayer periodic transmission structures with super-algebraic convergence.	8.2	Important neighbouring Fourier-Galerkin/multi-interface literature.
30	Hettlich, domain derivative of time-harmonic EM waves at interfaces, 2012. ⁵³	Fréchet/domain derivative for penetrable EM interfaces.	8.2	Direct theoretical relevance to dielectric geometry derivatives.

Rank	Citation / year	Short relevance summary	Relevance	Why important for novelty
31	Hao, Barnett, Martinsson & Young, high-order Nyström quadratures on smooth curves, 2011/2014. ⁷	Comparative analysis of Kress, Alpert, Kapur-Rokhlin and panel quadratures.	8.1	Useful for defining the nodal validation baseline.
32	Zhang & Zhang, simultaneous interface and buried-obstacle reconstruction, 2012. ⁵⁴	Newton iteration simultaneously reconstructs a penetrable interface and a buried obstacle from multifrequency data.	8.0	Strong prior art for nested/buried multi-interface inversion.
33	Kress, Yaman, Yapar & Akduman, impedance cylinder buried in a dielectric cylinder, 2009. ⁵⁵	Boundary-integral direct/inverse formulation for arbitrary-shaped nested dielectric/impedance cylinders.	8.0	Especially relevant to GPR/NDE-style buried configurations.
34	Cross-borehole inverse scattering using a boundary finite-element method, 1998. ⁵⁶	Shape reconstruction from borehole EM measurements using rigorous boundary-integral/finite-element direct solves.	7.8	Early geophysical/subsurface precedent.
35	Sugisaka, Harada & Hirayama, buried-defect reconstruction beneath a dielectric rough surface, 2022. ⁵⁷	Domain-boundary-integral hybrid inversion retaining multiple scattering.	7.8	Modern buried-target precedent close to GPR/NDE use cases.
36	Li et al., time-domain microwave imaging of a buried dielectric cylinder, 2010. ⁵⁸	Uses closed cubic-spline contour representation instead of trigonometric series.	7.7	Direct evidence that spline geometry is an established alternative in buried dielectric inversion.
37	Vlašić et al., implicit neural representation for mesh-free inverse obstacle scattering, 2022. ⁵⁹	Neural signed-distance/level-set geometry coupled to an implicit boundary-integral method and automatic differentiation.	7.5	Modern implicit-geometry competitor; relevant to topology flexibility.

Rank	Citation / year	Short relevance summary	Relevance	Why important for novelty
38	Mitzner, integral-equation scattering from finite-conductivity bodies, 1967. ⁶⁰	Classical coupled surface-current formulation for penetrable/ conductive electromagnetic bodies.	7.3	Historical EM surface-integral foundation.
39	Bulygin et al., Nyström method for Müller equations on dielectric bodies of revolution, 2015. ⁶¹	Nyström discretisation of Müller BIEs for dielectric resonators.	7.2	Additional confirmation of mature Müller+Nyström methodology.
40	Fang, Jiang & Su, fast Fourier-Galerkin BIE on torus-shaped surfaces, 2024. ¹⁴	Derives representation-matrix decay and truncates Fourier-Galerkin matrices to near-sparse form with proved convergence.	9.0 for proposed novelty	Not the same scattering problem, but possibly the most important conceptual precedent for the proposed modal-compressibility theorem.

The literature therefore suggests a very specific novelty hierarchy:

multi-object alone: low novelty
multi-material Müller alone: now low-to-moderate novelty
topology change alone: low novelty
Fourier-Galerkin Müller alone: low novelty
geometry-driven modal compression + derivatives + inversion: high potential
arbitrary-shape local-response reuse + inverse derivatives: high potential

The safest high-value research story is consequently **not “make the existing method more general” in the abstract**. It is:

Start from the explicit Laurent geometry and ask what that representation lets the forward and inverse operators do that generic BEM geometry cannot do cheaply.

If the first modal-compressibility experiment succeeds, that gives a clean mathematical/computational thesis: the **geometry spectrum predicts the scattering-operator coupling spectrum**, enabling an adaptive exact/spectral Müller solver and derivative engine. Multi-object and multi-material problems then become compelling demonstrations of that machinery rather than the novelty claim itself. If it fails, the work has still cheaply identified the better route: reusable local object response matrices and hierarchical/factorisation reuse, where the existing literature shows clear practical value but leaves room for an explicitly differentiable arbitrary-shape inverse-scattering architecture.

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